LING406_Lecture12_Notes

Course: LING406: Psycholinguistics
Title of Lecture: Lecture 12: Speech Perception II: Words
Date: 10/1/2026
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Section 1: A Very Old and Very General Model of Speech Perception

- The model from **Klatt (1979, Journal of Phonetics)** describes speech perception as a series of stages. It begins with the **speech waveform**, the analog signal that reaches the ear. It ends with **lexical hypotheses**, the candidate words the listener considers. Each stage takes the output of the previous stage and turns it into a more abstract representation.
    - The **peripheral auditory system**, which includes the cochlea, receives the waveform and performs a **spectral analysis**. This breaks the incoming sound into its frequency components.
    - The spectral analysis feeds the **acoustic property detectors**. Their outputs are tracked as detector outputs versus time, so they record how specific acoustic properties of the signal change from moment to moment.
    - The detector outputs feed the **phonetic feature detectors**. Their outputs are **feature probabilities versus time**. Each phonetic feature, such as whether a sound is voiced, receives a probability that changes across time instead of a fixed yes-or-no value.
    - The feature probabilities feed **segmental analysis**, which produces a **segmental feature matrix**. This matrix lists the features that make up each segment, where a segment is a phoneme.
    - The segmental feature matrix feeds the **lexical search**, which produces the lexical hypotheses.
- **Categorical perception** is the idea that somewhere between the analog signal entering through the cochlea and the activation of phonemes in segmental analysis, **information is lost**. Fine acoustic differences within a category are discarded, and what remains is the category itself.
    - This **loss of information is helpful**. Phoneme categories can be matched against stored word representations, which lets perception make contact with higher levels of representation.
- A natural assumption follows from this model. In order to identify words, a listener first has to identify their constituent phonemes. Two problems challenge this simple view: **cascaded processing** and **feedback**.

Section 2: A Familiar Voicing Continuum

- A **voicing continuum** is a series of speech sounds that changes in small, equal steps from one voicing category to another. Here the continuum runs from "peach" to "beach" in nine steps. The cue that changes is **voice onset time (VOT)**, the delay between the release of the consonant and the beginning of vocal fold vibration. Moving from "peach" toward "beach," the voice onset time becomes shorter.
    - The spectrograms plot frequency from 0 to 5000 Hz across 0.5 seconds. In the vowel portion, dark horizontal bands mark concentrations of energy at particular frequencies, and vertical striations mark the individual pulses of vocal fold vibration.
    - The "peach" endpoint has a longer interval before voicing begins than the "beach" endpoint. The steps in between have intermediate delays.
- The **identification function** plots how listeners label each of the nine steps. Steps 1 through 5 receive a value of 1, meaning they are consistently labeled as the "peach" end. Step 6 receives a value of about 0.33. Steps 7 through 9 receive a value of 0, meaning they are consistently labeled as "beach." The sharp drop between steps 5 and 7 is a step-like function rather than a gradual slope, and step 6 (circled in red) is the one stimulus that produces mixed labeling.

- The **discrimination function** plots how well listeners tell apart pairs of stimuli that are two steps apart: 1-3, 2-4, 3-5, 4-6, 5-7, 6-8, and 7-9. Discrimination is 0 for the pairs 1-3, 2-4, and 3-5. It reaches 1 for 4-6 and 5-7, falls to about 0.33 for 6-8, and returns to 0 for 7-9.
    - Every pair differs by the same two steps of voice onset time. Listeners distinguish the pairs that straddle the category boundary and fail to distinguish the pairs that fall inside a single category. This is the signature of categorical perception.

Section 3: How Does Categorical Perception Happen? A Threshold Model

- A **threshold model** proposes that activity is only passed forward from one level to the next when an activity threshold is crossed. Activity below the threshold stays at its own level and has no effect on the level above. In this kind of model, categorical perception arises at **phoneme identification**.
- The feature activation and phoneme activation panels make this concrete. Feature activation fluctuates over 100 time units, with peaks at about 1.05 and 1.2 and troughs at about −0.85 and −0.7. Dashed threshold lines sit at about +1.25 and −1.25. Activity above the upper threshold corresponds to **+voiced**, which maps to $/b/$. Activity below the lower threshold corresponds to **−voiced**, which maps to $/p/$.
    - The early peaks in feature activation stay below +1.25, so nothing is passed to the phoneme level. Phoneme activation remains flat at 0 for most of the time window.
    - Near the end, feature activation climbs to about 1.4 and crosses the upper threshold. Only then does the phoneme level register $/b/$, jumping to about 1.5.
    - The phoneme level therefore receives a categorical output, either $/b/$ or $/p/$, and none of the graded fluctuation that occurred in the feature activation underneath.

Section 4: How Does Categorical Perception Happen? A Cascaded Model

- A **cascaded model** proposes that activity is passed forward continuously over time from one level to the next. No threshold has to be crossed before the next level receives the activity. In this kind of model, categorical perception might happen later on, or might even be an artifact of the task structure.
- In the cascaded version of the same panels, the feature activation curve is identical, with the same peaks, troughs, and threshold lines. The phoneme activation curve now follows the same oscillating pattern. It rises to about 1.0, falls to about −0.9, rises again to about 1.1, falls to about −0.85, and finishes near 1.2.
    - The phoneme level carries the full graded pattern of feature activation, including the stretches where the evidence is weak or ambiguous.
    - A categorical response, such as the step-like identification function, would then be produced later in processing or by the demands of the task, because the phoneme level itself is not categorical.

Section 5: A Metaphor for Thresholded Activity: Making Furniture

- Making furniture is a metaphor for thresholded activity. The process has three layers. Each layer has its own input and output, and each layer works only after the previous layer has finished.
    - The **lumber mill** takes trees as input and produces lumber as output.
    - The **furniture factory** takes lumber as input and produces unassembled furniture as output.
    - The **IKEA customer** takes unassembled furniture as input and produces assembled furniture as output.
- Each layer of the system can only take a specific input that has been transformed **completely** by the previous layer. The furniture factory has to wait for the shipment of lumber before it can start manufacturing, and the customer has to wait for the unassembled furniture before assembly can begin. Many computer programs function this way.
    - A partially processed tree cannot be used by the factory. Activity does not pass forward until the previous layer has completed its transformation, which is the

same logic as a threshold that must be crossed before the next level receives anything.

Section 6: A Metaphor for Cascaded Processing: A Food Web

- A **food web** is a metaphor for cascaded processing. In a desert food web, desert plants are eaten by insects, small lizards, and kangaroo rats. Insects are eaten by scorpions and by large lizards. Small lizards are eaten by large lizards. Kangaroo rats are eaten by snakes, desert foxes, and hawks. Snakes, large lizards, and scorpions are in turn eaten by desert foxes and hawks. Many species connect to several others at once.
- The "levels" of a food web continuously interact with one another. A drought that reduces the number of plants immediately has an impact on the health of the kangaroo rats, which then impacts the snakes, and so on up through the web. No level has to finish changing before the next level is affected.
    - A food web also contains clear **feedback loops**. Adding more hawks changes the populations of snakes, lizards, and rats, and those changes in turn affect the hawks' own food supply.
- The question for speech perception is whether phoneme identification for word recognition works more like **thresholded** processing, as in the lumber mill, or **cascaded** processing, as in the food web.

Section 7: McMurray et al. (2003): Testing Identification in a Visual World Display

- **McMurray, Tanenhaus, Aslin, and Spivey (2003)** tested identification of a $/b/$ to $/p/$ continuum using a "visual world" display for **eye-tracking**. The study was published as "Probabilistic constraint satisfaction at the lexical/phonetic interface: Evidence for gradient effects of within-category VOT on lexical access" in the Journal of Psycholinguistic Research, volume 32, pages 77–97.
    - Each display contains four pictures: a **target** (the "pear"), a **competitor** (the "bear"), and two **filler** items (a "lamp" and a "ship"). Because "bear" and "pear" differ only in their first sound, the competitor is the other member of a minimal pair.
- Participants used the mouse to select the picture most consistent with what they heard, such as a "bear" or a "pear." The resulting identification function was typical and step-like.
    - The proportion of $/p/$ responses stays near 0 from 0 ms through about 15 ms of VOT, then rises sharply to about 0.95 at 20 ms. It stays at roughly 0.9 to 1.0 from 20 ms through 40 ms.
    - The **category boundary** was at 17.5 ± 0.83 ms of VOT.
    - Mouse responses therefore divide cleanly into two categories, the same way the identification function did in the "peach" to "beach" continuum.

Section 8: Eye Movements Reveal That Ambiguity Is Gradient

- Eye movements during the same task reveal that **ambiguity is gradient**. There are more looks to the competitor for stimuli close to the category boundary. Mouse responses are categorical, while eye movements record the competition between the two alternatives as it unfolds before the response is made.
- The fixation plots track the proportion of looks to the competitor over time since word onset, from 0 to 2000 ms. The left panel includes only trials where the response was "bear," so the competitor is the pear. These trials cover VOTs of 0, 5, 10, and 15 ms. The right panel includes only trials where the response was "pear," so the competitor is the bear. These trials cover VOTs of 20, 25, 30, 35, and 40 ms.
    - In both panels, looks to the competitor rise from baseline, peak at roughly 400 to 500 ms after word onset, and then decline.
    - Competitor looks are highest and last longest for the VOT nearest the boundary. In the left panel that is 15 ms, with a peak of about 0.15. In the right panel that is 20 ms, with a peak of about 0.13 and a slow decline that stays elevated past 1200 ms.
    - Values farther from the boundary, such as 0 ms on the left and 40 ms on the right, produce lower peaks and a faster return to baseline.

- The effect of VOT is primarily on the **duration of activation**. Stimuli nearer the boundary keep the competitor active for longer, which is a graded effect even though every one of these trials ends in the same categorical response.

Section 9: Early and Late Windows of Competitor Fixations

- The same data can be summarized by averaging fixations over two time windows: **early (300–1100 ms)**, plotted as a dotted line, and **late (1100–1900 ms)**, plotted as a solid line. The category boundary sits between 15 and 20 ms, and the lines break there because the response flips from "bear" to "pear."
- For "bear" responses (0 to 15 ms of VOT), looks to the pear competitor increase as VOT approaches the boundary.
  - In the early window, competitor fixations rise from about 0.075 at 0 ms to about 0.093 at 10 and 15 ms.
  - In the late window, competitor fixations rise from about 0.018 at 0 ms to about 0.043 at 15 ms. The late-window values more than double, which makes the gradient easier to see.
- For "pear" responses (20 to 40 ms of VOT), looks to the bear competitor decrease as VOT moves away from the boundary.
  - In the early window, competitor fixations start at about 0.105 at 20 ms, fall to about 0.085 at 25 and 30 ms, and reach a low of about 0.052 at 35 ms before rising slightly to about 0.067 at 40 ms.
  - In the late window, competitor fixations fall from about 0.053 at 20 ms to about 0.025 at 25 ms, and then drift down to about 0.015 at 40 ms.
- On both sides of the boundary, the stimuli closest to the boundary draw the most looks to the competitor. Stimuli that all produce the same response still differ in how much they activate the alternative word.

Section 10: Cascaded Activation From Features to Phonemes

- The finding is consistent with the idea that activation flows in a **cascaded** way from features to phonemes. The model in the figure has three layers. The **feature layer** receives a **pseudo-spectral input representation** and has rows labeled POW, VOC, DIFF, ACU, GRD, VOI, and BUR across a time axis from 0 to 44. The **phoneme layer** contains phoneme units such as $/b/$, $/t/$, and $/r/$, repeated at many starting times. The **word layer** contains word units, also repeated at many starting times.
  - Repeating each unit across time lets a feature, phoneme, or word begin at different points in an unfolding utterance.
  - Activation moves upward from the feature layer to the phoneme layer as the input arrives. Curved connections with filled endpoints link units within the same layer.
- Even stimuli that are identified as $/b/$ 100% of the time can vary in how strongly and unambiguously the "voiced" feature is activated. Two stimuli can both be labeled $/b/$ on every trial while differing in how much evidence for voicing they provide.
- If a stimulus is obviously +**bilabial** and +**stop**, this creates some momentary ambiguity between $/b/$ and $/p/$. These two phonemes share the bilabial place of articulation and the stop manner of articulation. They differ in voicing, so the voicing feature has to separate them.
  - That ambiguity is reflected in eye movements while the participant decides where to click. The competitor receives looks in proportion to how much of its supporting activation remains.

Section 11: Evidence That Partial Activation Can Reach All the Way to the Lexical Level

- **Andruski et al. (1994)** showed that **semantic priming** can occur even when the prime is a nonword. In a semantic priming task, a word related in meaning to the target speeds the response to that target. Here, a prime that is a slightly altered version of a real word still speeds responses to the word's semantic associate. "Beace" primes "war," and "ging" primes "queen."
  - The participants performed a **lexical decision task** in Experiment 1, judging whether each target was a real word.

- Prime word types were **unaltered**, **−1/3**, **−2/3**, and **unrelated**. The unaltered, −1/3, and −2/3 primes are semantically related to the target. Black bars mark "word → word" items, where the voiced counterpart of the prime is a real word, as in "pet → bet" or "bill" priming "drug." Shaded bars mark "word → nonword" items, where the voiced counterpart is a nonword, as in "king → ging."
    - For word → word items, mean reaction times are about 858 ms for unaltered primes, 857 ms for −1/3, and 888 ms for −2/3. The unrelated condition is slowest at about 965 ms.
    - For word → nonword items, mean reaction times are about 807 ms for unaltered primes, 820 ms for −1/3, and 815 ms for −2/3. The unrelated condition is again slowest at about 945 ms.
    - Reaction times in the altered conditions stay well below the unrelated baseline in both item types, so altered primes still facilitate their semantic associates.
- Lexical representations can therefore be activated even by incomplete phonetic evidence. Partial activation does not stop at the phoneme level and reaches the lexical level.

Section 12: Evidence for Cascaded Processing in Speech Perception

- The gradient pattern of looks in McMurray et al. (2003) is one piece of evidence for **cascaded** processing in speech perception. In a thresholded model, only a stimulus that crosses the threshold passes any information forward, and activity below the threshold is blocked from reaching the phoneme level, as the X on the arrow in the feature-to-phoneme panels indicates.
    - A thresholded model would predict the same pattern of looks to the competitor for all stimuli within a category, since none of their sub-threshold differences would reach the next level.
    - The number of looks to the other member of the minimal pair instead increases gradually as the stimulus approaches the category boundary. This gradual increase is hard to explain in a thresholded model and follows directly from cascaded activation.

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Section 13: Feedback From the Lexical Layer to the Phoneme Layer

- The three-layer model with a **feature layer**, a **phoneme layer**, and a **word layer** also incorporates **feedback** from the lexical layer to the phoneme layer. Arrows between the word layer and the phoneme layer point in both directions. The upward arrow carries activation from phonemes to the words they support, and the downward arrow carries activation from words back to their component phonemes. The arrow from the feature layer to the phoneme layer points only upward.
    - **Feedback** means that a higher level of representation sends activation back down to a lower level. In a model with only upward connections, the phoneme level would never be affected by what the word level is doing.
    - The word layer contains units written in phonetic notation, such as strings built from /b/, /r/, /i/, and /d/, each repeated at many starting times. The phoneme layer contains units such as /b/, /t/, and /r/, also repeated across time. Curved connections with filled endpoints link competing units inside a layer.

Section 14: Gisses and Kifts for Valentine's Day: The Ganong Effect

- The title of this topic is built from two nonwords. "Gisses" and "kifts" are nonwords that arise when the initial consonants of the real words "kisses" and "gifts" have their voicing swapped, so a listener who hears an ambiguous sound at the start of these items has a real word available for only one of the two interpretations.
- **Ganong (1980)** looked at how phoneme identification is impacted by word knowledge. He constructed word/nonword pairs such as "corpse/gorpse," in which one member of the pair is a real word and the other differs only in the voicing of the first consonant. He then produced continua spanning the relevant phoneme boundary in 5 ms steps.

    - In one kind of continuum, one endpoint is a word and the other is a nonword. In the opposite kind of continuum, the lexical status of the endpoints is reversed. A third, **neutral** condition provides a baseline.
    - The measure is the **proportion of voiced responses** at each point along the continuum. The horizontal axis runs from a short VOT through steps of −7.5 ms, −2.5 ms, +2.5 ms, and +7.5 ms around the boundary, out to a long VOT.
- One prediction comes from a late **correction** process. In this account, categorical perception functions as usual, and the phoneme is identified without any influence from the word level. Afterward, there is a **bias to report having heard a word**, so the listener reports whichever interpretation forms a real word.
    - Under this prediction, the bias applies equally at every point along the continuum. The **lexical effect**, which is the gap between the two curves, is a flat line at about 0.25.
    - Even the clearest stimuli would be affected. At the long VOT end, the continuum where the voiced interpretation is a word receives about 0.25 voiced responses even though the stimulus is clearly voiceless. At the short VOT end, the continuum where the voiced interpretation is a nonword receives only about 0.75 voiced responses even though the stimulus is clearly voiced.
    - In the middle of the continuum, the word-nonword curve falls from about 0.9 at −2.5 ms to about 0.45 at +2.5 ms, and the neutral curve falls from about 0.73 to about 0.27 across the same steps.

Section 15: A Prediction If Word Knowledge Shifts the Categorical Boundary

- A second prediction is that word knowledge works to **shift the categorical boundary**. In this account, lexical information changes where the dividing line between $/b/$ and $/p/$ falls. The curve for the continuum favoring the voiced interpretation as a word moves toward the voiceless end, and the curve for the opposite continuum moves toward the voiced end.
    - The curves still converge at both endpoints. At the short VOT end, all conditions give about 1.0 voiced responses, and at the long VOT end, all conditions give about 0.
    - The **lexical effect** is strongest for the items closest to the category boundary. It rises from about 0 at the short VOT end to about 0.1 at −7.5 ms, reaches about 0.38 at both −2.5 ms and +2.5 ms, drops to about 0.1 at +7.5 ms, and returns to about 0 at the long VOT end.
    - The effect peaks at the boundary because those stimuli are the ones for which lexical information can help resolve some ambiguity about the initial phoneme. A stimulus that is clearly voiced or clearly voiceless has no ambiguity left to resolve.

Section 16: The Results of Ganong (1980)

- The results are strikingly consistent with the second model. The lexical effect is small at the short VOT end, at about 0.02, and rises to about 0.12 at −7.5 ms. It peaks at about 0.32 at −2.5 ms, which is next to the boundary. It then falls to about 0.19 at +2.5 ms, about 0.12 at +7.5 ms, and about 0.09 at the long VOT end.
    - The word-nonword curve falls from about 0.95 voiced responses at −7.5 ms to about 0.67 at −2.5 ms, about 0.28 at +2.5 ms, and about 0.14 at +7.5 ms. It ends near 0.11 at the long VOT end.
    - The nonword-word curve falls more steeply, from about 0.83 at −7.5 ms to about 0.34 at −2.5 ms, about 0.09 at +2.5 ms, and about 0.02 at +7.5 ms.
    - The gap between the curves is therefore largest near the boundary and shrinks toward the endpoints, which is the pattern predicted when word knowledge shifts the categorical boundary and not the pattern predicted by a constant bias to report a word.
- Ganong raises a number of possibilities for how this could happen, focusing on different kinds of **guessing strategies**. Later models would emphasize the potential role of **feedback** in these shifts.

Section 17: A Simulation of the Ganong Effect in the TRACE Model

- The simulation follows an input in which the first segment is ambiguous between /b/ and /p/, followed by /l/, /\mathrm{^}/, and /g/. Four frames capture phoneme-level and word-level activation at successive points as the input unfolds. Word-level activation is plotted in the upper half of each frame on a scale up to 0.45 and above, and phoneme-level activation is plotted in the lower half on a scale up to 0.7 and above.
    - In the first frame, only the ambiguous initial segment has arrived. The phoneme units for /b/ and /p/ are both active at about 0.2, and word-level activity is low.
    - In the second frame, /l/ has arrived. Activation at the phoneme level now includes /l/ and /r/ alongside /b/, and a large cluster of word units is active at about 0.1.
    - In the third frame, /\mathrm{^}/ has arrived. The word unit for "plug" appears at about 0.28, and a cluster of other word units sits at roughly 0.1 to 0.18. "Blood" and "blush" are among these, because both begin with the sequence of /b/, /l/, and /\mathrm{^}/.
    - In the fourth frame, the final /g/ has arrived. The word unit for "plug" rises to the top of the scale, above 0.45. The competing words stay low, because a final /g/ is highly inconsistent with "blood" and "blush."
- As the input unfolds over time, activity accumulates for "plug," especially when the final /g/ arrives. No word beginning with /b/ and continuing with /l/, /\mathrm{^}/, and /g/ is available to compete, so the lexical evidence favors the interpretation of the ambiguous segment as /p/.
- This model explains word effects on phoneme category boundaries in terms of **top-down processing**. Top-down processing is influence flowing from higher-level representations, here words, down to lower-level representations, here phonemes. Specifically, activity from the lexical level can produce **feedback** to the phoneme level, producing greater activation for the phoneme consistent with a word.
    - The word "plug" gains activation as its sounds accumulate. That activation flows back down to the phoneme level and raises the activation of /p/ relative to /b/.
    - The same process accounts for the Ganong data. An ambiguous initial segment receives extra support for whichever phoneme completes a real word, which moves the boundary between the two categories.

Section 18: TRACE as a Version of the Interactive Activation Model

- The **Interactive Activation Network Model** (after **McClelland and Rumelhart, 1981**) is a model of written word recognition with three kinds of units. At the bottom are **feature units**, which correspond to simple line segments such as horizontal, vertical, and diagonal strokes. In the middle are **letter units**, including units for D, K, and R in the final position. At the top are **word units**, including WEAR, WORK, WORD, and WEAK.
    - Feature units send activation up to the letter units that contain those features. Letter units send activation up to the words that contain those letters. Connections also run back downward, and units at the same level are interconnected.
- A hypothetical set of features extracted on a trial in an experiment on word perception has parts of the letters covered by black patches. The visible strokes cover the beginning of the word clearly, and the final letter is partly hidden, so its visible features are compatible with more than one letter.
    - With incomplete feature information, several letter units and several word units become partially active at once. Activation at the word level feeds back to the letter level, so a letter that completes a plausible word receives support beyond what its visible features provide.
- The **TRACE** model is a version of the Interactive Activation model for written word recognition. Many of the principles are the same, including units at multiple levels, activation passing in both directions, and competition among units. A key issue in speech perception is the way the stimuli **unfold over time**. A printed word is available all at once, while a spoken word arrives one segment after another, which is why TRACE repeats its units across many starting times.

Section 19: Categorical Perception and the "Lexical-Phonetic Interface"

- Perception appears to be shaped by category boundaries. Given the same physical difference, it is more difficult to distinguish stimuli from the same category than stimuli from different categories. The evidence about cascaded activation and lexical feedback does not undermine that basic finding.
- **Ambiguity about phoneme identity does have an impact on word recognition.** Partial activation of $/b/$ and $/p/$ passes forward to the word level, which is why looks to the competitor in the eye-tracking data increase near the boundary and why altered primes still activate their semantic associates.
    - **Word-level information seems to have an impact on phoneme categorization.** The Ganong effect shifts the category boundary toward the interpretation that forms a real word, and the TRACE simulation produces the shift through feedback.
- The word recognition system therefore cannot simply wait for the phoneme recognition system to finish doing its job, as the lumber mill in the furniture metaphor waits for lumber. **Spoken word recognition seems to be more interactive than this.** Activity moves forward continuously and feedback moves backward, so the phoneme and word levels influence each other while the input is still arriving.

Section 20: Spoken Word Recognition Is Also Very Fast (Allopenna et al., 1994)

- **Allopenna et al. (1994, Journal of Memory and Language)** recorded eye movements as listeners heard a spoken word while viewing a grid of pictures. The display is a five-by-five grid containing a fixation cross in the center and several objects scattered across the squares. The objects include a flask, a speaker, a beetle, and a baby carriage, along with plain geometric shapes such as a triangle, a black square, a black circle, and a diamond.
    - The pictured objects set up four item types. The **referent** is the named target, a "beaker," which is the flask. The **cohort** is a word that begins with the same sounds as the target, a "beetle." The **rhyme** is a word that ends with the same sounds as the target, a "speaker." The **unrelated** item shares nothing with the target, a "carriage."
    - The measure is the **probability of fixating** each item type over time since target onset, in the full competitor condition of Experiment 1, averaged over all stimulus sets.
- Fixations to the referent begin to rise at about 200 ms after the onset of the target and continue climbing steadily until they level off near 0.95 at roughly 800 ms. The average target offset occurs at about 400 ms, and at that moment the probability of fixating the referent is only about 0.2. Eye movements are therefore directed toward the target before the word has finished.
    - **Cohort** fixations rise along with the referent fixations and are about equal to them until roughly 300 to 400 ms, because the beginning of "beaker" and the beginning of "beetle" are the same. Cohort fixations peak at about 0.18 at around 500 ms, then fall back to near 0 by about 800 ms once the ending of the word rules the cohort out.
    - **Rhyme** fixations rise somewhat later and stay lower. They peak at about 0.1 at around 500 to 600 ms and decline slowly, and they are still just above baseline near 1000 ms. The rhyme competitor receives some activation even though it differs from the target at the beginning, because the later part of the word matches.
    - **Unrelated** fixations remain near 0 throughout, with only a very small rise at later times.

Section 21: TRACE Predictions for the Eye-Movement Data

- The simulation plots the fixation probabilities predicted by TRACE alongside the observed fixation probabilities. In the cohort comparison, TRACE and the data both show the referent rising to about 0.95 by 800 to 1000 ms. The predicted curve rises slightly earlier and more steeply than the observed curve, and the two converge once they level off. The cohort curve in the model peaks at about 0.17 at around 500 ms, closely matching the observed cohort peak, and both fall back toward 0 by about 800 ms.
    - In the rhyme comparison, the referent curves follow the same pattern. The predicted rhyme curve and the observed rhyme curve both peak at about 0.1 at

around 500 to 600 ms and decline slowly afterward.
    - A model with activation at multiple levels and competition among words produces both the early overlap between the referent and the cohort and the later, smaller rise of the rhyme.
- The simulation is **granular**. Eye movements give an idea about the changes in "activity" for different competitors over time, so cognitive processing can be observed before a final response is issued. Each competitor's activation rises and falls on a timescale of hundreds of milliseconds, and those changes appear in the fixation record long before a participant clicks or answers.
    - Within-category differences in how good an exemplar is also produce **graded effects of category goodness** in the categorical perception task. The graded pattern in the eye movements follows from the same kind of continuous activation.